# Project Report - Day 1

## Title

TASK-01: Building a Resilient Pipeline for Live Weather Telemetry

## Objective

To fetch weather data from OpenWeatherMap API, process it, and store it in a structured JSON format.

## Libraries Used

- requests
- json
- os
- dotenv
- datetime

## Tools Used

- Vs Code
-Microsoft Word
-GitHub

## Errors

1. **Incorrect API URL**
   At first, I used a normal website link instead of the actual API endpoint. Because of this, the response was in HTML format instead of JSON, which caused a JSONDecodeError.

2. **Typing and Syntax Errors**
   There were small mistakes like:
   Wrong variable name (respoonse instead of response)
   Incorrect condition (status code instead of response.status_code)
   Typo in URL (ci(ty} instead of city).

3. **Incorrect Use of Loop**
   Some parts of the code like data extraction and storing results were written outside the loop. Because of this, only one city's data was saved instead of all cities.

## Solutions

1. **Correct API Endpoint**
   I used the proper API URL:
   http://api.openweathermap.org/data/2.5/weather

2. **Fixed Syntax Errors**
   All typos and incorrect variable names were corrected to ensure smooth execution.

3. **Implemented Error Handling**
   A try-except block was added, and status codes were checked to handle:
   Invalid city names (404)
   Invalid API key (401)

## Conclusion

This task helped in understanding API integration, data transformation, and error handling. It improved practical knowledge of Python and real-world data processing.